

Engineering Design Report

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£30.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction
[E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	FAIL	No regulations listed	0 standards
3. Cost Ceiling Specified	PASS	Target provided	£30.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	10 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements [A]

1.1 Mission & Use Case

Mission

Not computed

Use Case

Not computed

1.2 Market Context & Timing [B]

Not computed

Not computed

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Not computed

Product	Ecodan CAHV-R450YA-HP (Hydronic Zubadan commercial range)
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Pricing	Industry sources suggest list pricing in the range of £8,000–£14,000 for units in the 28–40kW class; verification required
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Tech Specs	Not computed
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STRENGTHS

Extremely strong brand recognition and installer trust in UK commercial market, Comprehensive MCS certification and UK support infrastructure, Broad operating range including low-ambient performance (Zubadan technology), Extensive distributor and service network

WEAKNESSES

R32 refrigerant faces medium-term F-Gas regulatory pressure, Significantly higher price points than target product, Product range complexity can be a barrier for smaller installers, Not optimised specifically for the sub-35kW small commercial sweet spot

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Daikin Altherma 3 R (commercial/high-capacity variants, e.g. ETVX series)
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Pricing	Published trade price indications suggest commercial Altherma variants in this output range are typically priced above £7,000–£12,000 list; verification required
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Tech Specs	Not computed
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STRENGTHS

Market-leading brand with very high installer familiarity, Robust MCS certification portfolio, Strong ErP ratings supporting BUS eligibility, Good parts and technical support ecosystem in UK

WEAKNESSES

R32 refrigerant carries long-term regulatory uncertainty under F-Gas phase-down, Premium pricing relative to proposed target, Commercial range product updates to R290 are anticipated but not yet widely deployed in this output class, Some installer feedback (industry forums) cites commissioning complexity

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	aroTHERM plus (commercial configurations, VWL 125/5 AS and similar)
Pricing	Industry data suggests UK trade pricing in the £4,500–£8,000 range for mid-capacity models; commercial variants may be priced higher; verification required
Tech Specs	Not computed

STRENGTHS

Already uses R290, providing direct regulatory alignment, MCS certified in the UK, Strong brand presence and installer training programme, Hydronic architecture well-matched to UK wet system retrofit

WEAKNESSES

Achieving 30kW output requires multi-unit manifolding, adding complexity and cost, No native single-unit commercial product at 30kW in the R290 aroTHERM line (as of available information; verification required), Residential-focused engineering may not fully address small commercial duty cycles and controls requirements, Manifolded installations complicate MCS compliance pathway and commissioning

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	NIBE F2120 (air-source split heat pump, 8–20kW range) and associated commercial configurations
Pricing	Industry sources suggest NIBE commercial split units in relevant capacities are priced in the £5,000–£9,000 range at trade; verification required
Tech Specs	Not computed

STRENGTHS

Reputation for reliability and Scandinavian engineering quality, Strong UK commercial installer base and technical support, Hydronic architecture well-suited to UK commercial retrofit, Good integration with building management systems

WEAKNESSES

R410A refrigerant carries significant F-Gas phase-down risk, Maximum single-unit output below 30kW target requires cascading for the target application, Brand is stronger in residential and light commercial than in the 30kW commercial segment, R290 transition timeline for commercial range not yet publicly confirmed (verification required)

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Midea M-Thermal commercial air-to-water heat pump (various OEM-badged variants in 25–35kW class)
Pricing	Trade sources suggest OEM-badged Midea commercial units in this class may be available at £2,500–£4,500 depending on specification and channel; verification required
Tech Specs	Not computed

STRENGTHS

Highly competitive unit cost, potentially at or below the target price point, Broad output range and configuration flexibility, Improving UK market presence through OEM and distribution partnerships

WEAKNESSES

MCS certification status for specific commercial models requires case-by-case verification; not all variants are MCS certified, R32 refrigerant faces F-Gas headwinds, Brand credibility and after-sales support perceived as weaker by UK commercial installers (industry commentary suggests; verification required), Technical documentation and commissioning support quality variable across OEM channels, No confirmed R290 commercial split unit in this class in UK market (verification required)

DIFFERENTIATION ANGLE

Not computed

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£30.00
Max Mass	Not computed

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2. Regulatory & Compliance

No regulatory standards identified for this brief.

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	hardware_product

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
compressor_drive_assembly	2236 × 2236 × 2400	5	floor
evaporator_coil_airflow_mod- ule	2236 × 2236 × 2400	5	floor
condenser_heat_ex- change_module	2236 × 2236 × 2400	5	floor
refrigerant_expansion_con- trol_valve_assembly	2236 × 2236 × 2400	5	floor
defrost_condensate_manage- ment_module	2236 × 2236 × 2400	5	floor
outdoor_chassis_enclo- sure_assembly	2236 × 2236 × 2400	5	floor
atex_safety_sensor_monitor- ing_module	2236 × 2236 × 2400	5	floor
outdoor_power_inverter_elec- tronics_module	2236 × 2236 × 2400	5	floor
indoor_hydrobox_pump_cir- culation_module	2236 × 2236 × 2400	5	floor

indoor_hydrobox_control_bms_interface	2236 × 2236 × 2400	5	floor
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Conflicts

- Total module footprint (50.0m2) exceeds envelope budget (21.3m2)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 Compressor & Drive Assembly [D]

Maturity: draft

Purpose

Compresses low-pressure R290 vapour into high-pressure, high-temperature gas to drive the thermodynamic cycle.

Why It Matters

Not computed

Technical Description

The assembly incorporates a 3-phase, ATEX-rated inverter scroll or twin-rotary compressor, sourced from Copeland, Danfoss, or GMCC, hermetically sealed to contain A3-classified propane refrigerant. The terminal box is intrinsically safe, rated to Ex ia IIC T4 or equivalent, preventing ignition of any leaked propane vapour at concentrations above 2.1% v/v LEL. Four elastomeric anti-vibration mounts, each rated to 150 kg static load, isolate low-frequency mechanical vibration in the 20–80 Hz range from the structural chassis. An integrated suction accumulator, with a minimum 0.5-litre liquid separation volume, prevents liquid R290 slugging the scroll plates during defrost cycle pull-down. The assembly is enclosed in a 25 mm closed-cell acoustic insulation jacket, reducing radiated sound power by approximately 6 dB(A) to meet the 42 dB(A) boundary noise limit under UK Permitted Development rights.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Evaporator Coil & Airflow Module [D]

Maturity: draft

Purpose

Absorbs ambient thermal energy from outdoor air to vaporise the liquid R290 refrigerant.

Why It Matters

Not computed

Technical Description

The coil is constructed from 9.52 mm outer-diameter copper tubes, mechanically expanded into 0.1 mm thick aluminium fins coated with a hydrophilic surface treatment. The hydrophilic coating reduces water droplet contact angle to below 10°, accelerating condensate shedding and delaying frost bridging at ambient temperatures between 0°C and -5°C by approximately 15 minutes compared with uncoated fins. Two vertically stacked, variable-speed Electronically Commutated axial fans, sourced from ebm-papst or Ziehl-Abegg with EC motor efficiency above 70%, draw air uniformly across the full coil face. Each fan is mounted within an aerodynamically profiled GRP shroud, reducing tip-vortex losses and maintaining measured sound power levels between 60 dB(A) at minimum speed and 68 dB(A) at full load. Protective polypropylene finger grilles prevent debris ingestion and maintain a minimum 20 mm clearance from blade tips.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 Condenser Heat Exchange Module [D]

Maturity: draft

Purpose

Transfers heat from the high-temperature R290 refrigerant gas into the secondary hydronic heating circuit.

Why It Matters

Not computed

Technical Description

The module centres on a brazed plate heat exchanger (BPHE) manufactured from 316-grade stainless steel with copper brazing, typically supplied by SWEP or Alfa Laval, providing a minimum effective heat transfer area of 0.8 m² per 10 kW of rated capacity. The BPHE is located entirely within the outdoor unit to comply with EN 378-1:2016 Annex C requirements, ensuring no R290 pipework penetrates the building envelope. A 25 mm layer of closed-cell elastomeric foam insulation, such as Armaflex HT, wraps the exchanger body, limiting heat loss to less than 2% of rated output at -10°C ambient. A water-side magnetic flow switch, set to a minimum trip threshold of 0.8 l/s, and dual high-pressure refrigerant cut-off switches, set at 28 bar gauge, prevent over-pressurisation if water flow falls below the safe operating envelope.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Refrigerant Expansion & Control Valve Assembly [D]

Maturity: draft

Purpose

Precisely meters liquid refrigerant flow to reduce its pressure and directs flow paths for heating or active defrost modes.

Why It Matters

Not computed

Technical Description

The sub-assembly integrates a Carel E2V-series Electronic Expansion Valve (EEV) driven by a 200-step stepper motor, providing flow resolution to within 0.5% of full stroke at operating pressures up to 30 bar. A heavy-duty 4-way reversing valve redirects R290 flow within 3 seconds to switch the system between heating and active hot-gas defrost modes. All copper pipework connections are factory-brazed using phosphor-bronze filler to BS EN 14324, minimising mechanical joints to zero in the refrigerant circuit. Every brazed joint undergoes 100% helium leak testing at 30 bar before dispatch, with an acceptance criterion of less than 1×10^{-6} mbar·l/s. Dual piezo-resistive pressure transducers on the high-side (0–40 bar range) and low-side (0–20 bar range) supply real-time telemetry to the inverter control board, maintaining dynamic load balance across the full 30 kW operating range. An inline solid-core filter drier, rated to a minimum 3 g moisture absorption capacity, protects the EEV needle and seat from particulate contamination.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 Defrost Condensate Management Module [D]

Maturity: draft

Purpose

Collects, routes, and prevents the freezing of condensate shed from the evaporator coil during and after defrost cycles.

Why It Matters

Not computed

Technical Description

The drain pan is fabricated from 1.5 mm thick e-coated galvanised steel or 304 stainless steel, formed with a minimum 3° slope across its full 600 mm width to direct condensate to the central 32 mm nominal bore drain port. For continuous operation at ambient temperatures between -15°C and -20°C, an ATEX-rated trace heating cable, rated at 10 W/m, is bonded along the full length of all drain channels and directly to the drain port outlet. The cable is controlled by a dedicated NTC thermistor embedded within the pan body, switching on at 2°C and off at 5°C to minimise electrical energy consumption. The 32 mm drain port accepts a push-fit connection to standard commercial soakaway pipework in accordance with BS EN 1253.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Outdoor Chassis & Enclosure Assembly [D]

Maturity: draft

Purpose

Provides structural rigidity, acoustic attenuation, and safe, code-compliant housing for all outdoor components.

Why It Matters

Not computed

Technical Description

The chassis frame is welded from 2 mm thick galvanised steel sections and finished with a zinc-rich e-coat primer of minimum 60 µm dry film thickness, followed by a polyester powder topcoat of 80 µm, providing salt-spray corrosion resistance exceeding 1,000 hours to BS EN ISO 9227. Interior panel faces are lined with 20 mm thick melamine-foam acoustic deadening material, contributing approximately 8 dB(A) insertion loss at compressor running frequencies, critical for compliance with the 42 dB(A) Permitted Development noise boundary limit. Specific to R290 installations, lower panel sections are engineered with 200 mm² per linear metre of open-area venting louvers positioned within 100 mm of the base, allowing propane vapour with a relative density of 1.52 to air to dissipate to atmosphere before accumulating to its LEL of 2.1% v/v.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 ATEX Safety & Sensor Monitoring Module [D]

Maturity: draft

Purpose

Detects R290 vapour concentrations above safe thresholds and triggers immediate fail-safe system lockout and forced ventilation.

Why It Matters

Not computed

Technical Description

Two or more factory-calibrated infrared or pellistor-type R290 leak detectors are mounted at the lowest internal chassis points, within 100 mm of the base plate, to intercept propane vapour accumulating at floor level before concentrations rise. Each sensor is calibrated to a detection threshold of 10% LEL, equivalent to 0.21% v/v R290 in air, with a response time of less than 30 seconds to a step-change in concentration. On alarm, dedicated solid-state safety relays, independent of the main processor board, de-energise all spark-risk loads including the compressor and EEV within 500 ms. Simultaneously, ATEX-rated EC fans are commanded to maximum RPM by the relay logic, purging the 0.15 m³ internal chassis volume at a minimum of 10 air changes per minute to reduce vapour concentration below 10% LEL. Volt-free alarm contacts rated at 24 V DC / 1 A connect to external warning beacons or BMS dry-contact inputs.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 Outdoor Power & Inverter Electronics Module

[D]

Maturity: draft

Purpose

Conditions 3-phase grid power and dynamically varies compressor and fan speeds across the full operating range.

Why It Matters

Not computed

Technical Description

The module houses a 400 V, 3-phase Variable Frequency Drive (VFD) printed circuit board assembly capable of managing a continuous electrical input of approximately 8.6 kW, with a peak overload rating of 120% for 60 seconds. The electronics assembly is mounted within an IP65-rated sealed aluminium sub-enclosure, positioned at the highest internal point of the chassis, at least 300 mm above the base plate, to remain above any propane vapour layer. A finned aluminium heat sink of minimum 0.25 m² exposed surface area protrudes from the sealed enclosure directly into the evaporator fan airstream, dissipating IGBT switching losses estimated at 2–3% of throughput power, equivalent to 170–260 W at full load. An EMC/RFI filter assembly attenuates conducted emissions to comply with EN 61000-3-2 Class A limits, meeting CE and UKCA marking requirements for 3-phase inverter equipment.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

- [LLM Architecture] Decomposition Logic
- [LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Indoor Hydrobox - Pump & Circulation Module ^[D]

Maturity: draft

Purpose

Distributes heated water from the outdoor heat exchanger to the building's central heating circuit.

Why It Matters

Not computed

Technical Description

The module is installed indoors and physically separated from all R290 refrigerant pipework, containing a variable-speed hydronic circulation pump such as the Grundfos MAGNA3 or Wilo Stratos, sized to deliver 1.4–1.7 l/s at a head of 4–6 m WG across 28–35 mm BSP pipework connections. A pre-plumbed low-loss header, minimum 100 mm bore, hydraulically decouples the primary heat pump circuit from the secondary building distribution circuit, preventing flow-rate mismatch across the outdoor BPHE. A 12-litre closed expansion vessel, pre-charged to 1.0 bar, accommodates system water expansion from cold-fill at 10°C to maximum operating temperature of 75°C without exceeding the 3 bar set point of the pressure relief valve. A full-bore magnetic dirt separator, rated for nominal 28 mm pipework, captures ferrous magnetite particles above 10 µm before they enter the outdoor BPHE plate channels.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.10 Indoor Hydrobox - Control & BMS Interface

[D]

Maturity: draft

Purpose

The primary microprocessor controller managing thermodynamic algorithms, temperature schedules, and external communications.

Why It Matters

Not computed

Technical Description

The module contains a main logic controller, typically supplied by Carel, Siemens, or Danfoss, operating on a 32-bit processor with a minimum 4 MB non-volatile programme memory for retention during power interruptions of up to 72 hours. A 7-inch resistive or capacitive LCD touchscreen interface allows the operator to set leaving water temperatures between 25°C and 75°C and configure 7-day heating schedules with 30-minute resolution. A weather compensation algorithm calculates target flow temperature using a configurable heating curve with a slope adjustable between 0.3 and 2.5 K per degree of ambient change, drawing ambient data from the outdoor chassis thermistor at 60-second intervals. Native Modbus RTU RS-485 and BACnet IP interfaces, both configured to standard object models, allow facilities managers to integrate the unit into master BMS platforms without gateway hardware. Cascade logic terminals support sequencing of up to 3 identical units, extending the combined output range to approximately 90 kW.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
Compressor & Drive Assembly	Internal electrical short igniting R290 mixture.	Not computed	Not computed	5×1=5	Hermetically sealed compressor vessel and exact ATEX certification for the terminal box.
Compressor & Drive Assembly	Liquid slugging causing catastrophic mechanical rupture.	Not computed	Not computed	4×2=8	Oversized suction accumulator and active superheat control via the expansion valve.
Evaporator Coil & Air-flow Module	Fan blade mechanical failure causing severe imbalance and enclosure damage.	Not computed	Not computed	3×2=6	Use of high-strength glass-reinforced plastic (GRP) or cast aluminum blades with continuous RPM anomaly monitoring.
Evaporator Coil & Air-flow Module	Severe internal frost rupture of copper tubes.	Not computed	Not computed	4×2=8	Dual parameter (time/temperature) defrost logic and regular coil shedding cycles.
Condenser Heat Exchange Module	Internal plate rupture mixing high-pressure R290 gas into the circulating water.	Not computed	Not computed	5×1=5	Use of heavily fatigue-tested PED-compliant BPHEs and rigorous external pump proving before compressor start.
Condenser Heat Exchange Module	Heat exchanger freezing solid leading to burst casing during power outages.	Not computed	Not computed	4×3=12	Mandatory integration of anti-freeze valves or dosing commercial circuits with glycol.
Refrigerant Expansion & Control Valve Assembly	Micro-leak of R290 gas from a stressed brazed joint.	Not computed	Not computed	4×2=8	Robust vibration loops in piping design and strict avoidance of unsealed mechanical joints.

Refrigerant Expansion & Control Valve Assembly	System over-pressurization leading to pipe burst.	Not computed	Not computed	4×1=4	Hard-brazed, redundant mechanical hi-pressure safety cut-out switches (PED mandated).
Defrost Condensate Management Module	Expanding ice physically crushing the bottom refrigerant loop and causing an R290 leak.	Not computed	Not computed	4×3=12	Redundant trace heating linked to ambient sensors, pan designed without mechanical traps.
Defrost Condensate Management Module	Electrical fault in the heater cable in a potentially damp/gaseous environment.	Not computed	Not computed	3×2=6	Low-voltage or ATEX-rated fully encapsulated cable elements, RCD protected.
Outdoor Chassis & Enclosure Assembly	Gas pooling within the chassis reaching the 2.1% explosion threshold.	Not computed	Not computed	5×2=10	Bottom-louvered 'open enclosure' architecture to allow continuous heavy-gas escape.
Outdoor Chassis & Enclosure Assembly	Enclosure toppling during extreme wind events.	Not computed	Not computed	3×2=6	Low center of gravity design and robust structural mounting rail provisions.
ATEX Safety & Sensor Monitoring Module	Leak goes undetected leading to an uncontrolled external ignition.	Not computed	Not computed	5×1=5	Use of dual, redundant LEL sensors with fail-safe self-diagnostic checks.
ATEX Safety & Sensor Monitoring Module	False positive shuts down heating to a commercial space during extreme winter.	Not computed	Not computed	2×3=6	Intelligent pre-alarm thresholds and direct BMS alerting for facilities maintenance.
Outdoor Power & Inverter Electronics Module	Arcing relays within the board igniting ambient propane gas.	Not computed	Not computed	5×1=5	Sealing electronics in ATEX-verified IP65 positive-pressure or hermetic boxes.
Outdoor Power & Inverter Electronics Module	High starting current violating UK DNO network limitations.	Not computed	Not computed	3×3=9	Strict firmware enforcement of slow-ramp soft start profiling.

Indoor Hydrobox - Pump & Circulation Module	Insufficient flow causing outdoor BPHE freeze and internal rupture.	Not computed	Not computed	5×2=10	Continuous flow switch monitoring interlocked directly to the compressor run circuit.
Indoor Hydrobox - Pump & Circulation Module	Over-pressurization of internal water piping.	Not computed	Not computed	3×2=6	Redundant PRVs terminating safely to a tundish.
Indoor Hydrobox - Control & BMS Interface	Total loss of communication leads to uncontrolled runaway heating.	Not computed	Not computed	3×2=6	Hardware 'watchdog' timers in the outdoor unit that auto-shutdown upon loss of indoor heartbeat signals.
Indoor Hydrobox - Control & BMS Interface	Unauthorized tampering changes critical heating limits creating a scalding risk (DHW >65°C).	Not computed	Not computed	4×3=12	Deep installer-level passcode protection and specific maximum set-point limits in software.

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	10 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success